

## Lecture 09: Modular Arithmetic

### 1 Quick Followup from Last Week

**Proposition 1.** For all integers  $a$  and  $b$ , the common divisors of  $a$  and  $b$  are precisely the common divisors of  $a$  and  $b - a$ .

*Proof.* Suppose  $d$  is a common divisor of  $a$  and  $b$ . Let  $x, y$  be integers such that  $dx = a$  and  $dy = b$ . Then  $d(y - x) = b - a$ , so  $d$  is also a common divisor of  $a$  and  $b - a$ .

Conversely, suppose  $d$  is a common divisor of  $a$  and  $b - a$ . Let  $x, y$  be integers such that  $dx = a$  and  $dz = b - a$ . Then  $d(x + z) = b$ , so  $d$  is also a common divisor of  $a$  and  $b$ .  $\square$

**Theorem 2** (Bezout's Identity + The Pulverizer). For any integers  $a, b$ , there exist integers  $s, t$  such that  $\gcd(a, b) = as + bt$ . We can compute  $s, t$  from  $a, b$  using the Pulverizer.

**Corollary 3.** A number can be written as an i.l.c. of  $a, b$  IFF it is a multiple of  $\gcd(a, b)$ .

*Proof.* Let  $g = \gcd(a, b)$ . Every i.l.c. of  $a, b$  is divisible by  $g$ . Conversely, we know  $g = sa + tb$  for some  $s, t$  by Bezout, so every multiple of  $g$  (say  $kg$ ) can be written as  $(ks)a + (kt)b$  and is therefore an i.l.c. of  $a, b$ .  $\square$

### 2 Towards Modular Arithmetic

- What is even+odd? (odd)
- What is the last digit of  $357 \times 994$ ? (8, b/c  $7 \times 4 = 28$ )
- It is currently 3pm. What time will it be in 49 hours? (4pm, b/c 49h is 1h more than 2 full days)
- Today is Tuesday. What day of the week will it be in 10 days? (Friday, b/c 10 days is 3 more than a week.)
- What day of the week will it be 100 days from now? What computation do you need to do? ( $\text{rem}(100, 7) = 2$ , so Tues+2=Thurs)

These are all familiar examples of Modular Arithmetic. When working modulo  $n$ , the theme is “ignore multiples of  $n$ , just focus on remainders”

Even/Odd: remainder when dividing by 2. Weekday: remainder when dividing by 7. Last digit: remainder when dividing by 10. Hour: remainder when dividing by 12 or 24 (if we care about am/pm).

$$\begin{aligned} dx &= a \\ dy &= b \end{aligned}$$

$$\begin{aligned} dy - dx &= b - a \\ \text{iff } d(y - x) &= b - a \\ &= b - a \end{aligned}$$

$$\begin{aligned} dx &= a \\ dy &= b - a \\ dx + dy &= a + (b - a) \\ &= d(y + x) = b \end{aligned}$$

Often called clock arithmetic, because we're familiar with ignoring multiples of 12 or 24 when telling time.

**Definition 1.** We say  $a \equiv_n b$  (pronounced "a is congruent to b mod n") IFF  $n \mid a - b$ .

Note: more standard notation for  $a \equiv_n b$  is  $a \equiv b \pmod n$  or  $a \equiv b \pmod n$ . However, this notation can invite confusion (explained later), so we suggest sticking to the  $a \equiv_n b$  notation until you are familiar with modular arithmetic.

We'd like to consider  $a$  and  $b$  "the same" when their difference is a multiple of  $n$ .

For example, there are only 5 different "values" when looking mod 5:

$$[0] = \{\dots, -10, -5, 0, 5, 10, \dots\}$$

$$[1] = \{\dots, -9, -4, 1, 6, 11, \dots\}$$

$$[2] = \{\dots, -8, -3, 2, 7, 12, \dots\}$$

$$[3] = \{\dots, -7, -2, 3, 8, 13, \dots\}$$

$$[4] = \{\dots, -6, -1, 4, 9, 14, \dots\}$$

For a general number  $k$ , which group will it belong to? Just look at  $k \bmod 5$ . If we write  $k = 5q + r$ , then  $k \equiv_5 r$ , because  $k - r = 5q$ .

Recall the Division Theorem:

**Theorem 4.** For all pairs of integers  $n, d$  with  $d > 0$ , there exists a unique pair of integers  $q, r$  where  $n = qd + r$  and  $0 \leq r < d$ . The number  $q = n \operatorname{div} d$  is the quotient, and  $r = n \operatorname{rem} d$  is the remainder.

$\Rightarrow$  side.

When working mod  $n$ , a number  $k$  is always congruent to its remainder (sometimes called its residue): if  $k = nq + r$ , then  $n \mid nq = k - r$ , so  $k \equiv_n r$ . Claim: the  $n$  remainders  $0, 1, \dots, n-1$  represent all the possible "groupings" (called residue classes or equivalence classes) mod  $n$ .

**Theorem 5.**  $a \equiv_n b$  if and only if  $(a \bmod n) = (b \bmod n)$ .

*Proof.* If  $(a \bmod n) = (b \bmod n) = r$  then  $a = nq + r$  and  $b = nq' + r$  for some  $q, q'$ . So  $a - b = n(q - q')$  which is a multiple of  $n$ , so  $a \equiv_n b$ .

Conversely, suppose  $a \equiv_n b$ , so  $a - b = nk$  for some  $k$ . Write  $b = qn + r$  where  $0 \leq r \leq n-1$ , so  $r = (b \bmod n)$ . Then  $a = b + nk = (q + k)n + r$ . Since  $0 \leq r \leq n-1$ ,  $k + q$  and  $r$  are the unique values guaranteed by the Division Theorem, i.e.,  $r$  also equals  $a \bmod n$ .  $\square$

So the  $n$  different remainders when dividing by  $n$  divide  $\mathbb{N}$  into  $n$  different groups, identified by their remainders. Can think of  $0, 1, \dots, n-1$  as the only possible values mod  $n$ , and all other numbers are congruent to one of these.

### 3 Interlude: Confusing Notation

#### 3.1 Remainder

Remainder can be notated as  $a \text{ rem } n$  aka  $\text{rem}(a, n)$  aka  $a \bmod n$ . Recall that  $n$  is always positive, but  $a$  can be pos or neg.

Many languages have the modulo operator  $a \% n$  which generally behaves like our  $\text{rem}$ , but not always!! By our def,  $a \text{ rem } n$  is always nonnegative, even when  $a$  is negative:  $(-43 \% 10) = 7$ . Python and Mathematica agree with us. But many *other* languages think negative  $a$  values should have negative remainders:  $(-43 \% 10) = -3$ . Javascript and C/C++, for example. And some have both, with two different names, e.g., CoffeeScript ( $\%$  vs  $\% \%$ ), Lisp ( $\text{mod}$  vs  $\text{rem}$ ), Fortran ( $\text{mod}$  vs  $\text{modulo}$ ), Haskell ( $\text{mod}$  vs  $\text{rem}$ ).

For this class, any version of remainder we use will always mean the nonnegative one.

Similarly,  $a // n$  is commonly used programming notation for integer division, but languages disagree on which way to round. We always round *down*.

#### 3.2 Two meanings for mod

Confusing notation:  $a \bmod n$  is commonly used for  $\text{rem}(a, n)$ . Confusing! What does  $a = b \bmod n$  mean? Does it mean  $a \equiv b \bmod n$ ? Or does it mean  $a = (b \bmod n)$ , i.e.,  $a = b \text{ rem } n$ ?

Difference:  $a \bmod n$  is a function, with a single definite value, namely  $a \text{ rem } n$ . Always between 0 and  $n - 1$ .

But  $a \equiv b \bmod n$  is a relationship between two quantities. Neither needs to be between 0 and  $n - 1$ . E.g.,  $12 \equiv 17 \bmod 5$  is a true statement. Their remainders are  $(12 \bmod 5) = 2$  and  $(17 \bmod 5) = 2$ .

### 4 Putting the Arithmetic in Modular Arithmetic

The simple statement even + odd = odd says something profound: “no matter which even number and odd number we add, the result is always odd”. This generalizes: “if we pick *any* number  $a \equiv_5 3$  and *any* number  $b \equiv_5 4$ , adding them will always produce a number  $a + b \equiv_5 7$ . (Could also write this as  $a + b \equiv_5 2$ .)

**Theorem 6.** If  $a \equiv_n b$ , then for any  $c$ ,

1.  $a + c \equiv_n b + c$ ,
2.  $ac \equiv_n bc$ ,
3.  $a - c \equiv_n b - c$ , and
4.  $c - a \equiv_n c - b$ .

*Proof.* By definition of  $\equiv_n$ ,  $n \mid a - b$ .

1.  $(a + c) - (b + c) = a - b$ , so  $n \mid (a + c) - (b + c)$ . Therefore,  $a + c \equiv_n b + c$ .
2.  $ac - bc = (a - b)c$ , a multiple of  $a - b$ , so  $n \mid ac - bc$ . Therefore,  $ac \equiv_n bc$ .
3.  $(a - c) - (b - c) = a - b$ , so  $n \mid (a - c) - (b - c)$ . Therefore,  $a - c \equiv_n b - c$ .
4.  $(c - a) - (c - b) = b - a$ , a multiple of  $a - b$ , so  $n \mid b - a$ . Therefore,  $c - a \equiv_n c - b$ .

□

When adding or multiplying or subtracting, can replace  $a$  by anything it is congruent to mod  $n$ , without changing the result mod  $n$ .

True for the **base** of exponents as well:

**Theorem 7.** If  $x \equiv_n y$ , then for any  $k \geq 1$ ,  $x^k \equiv_n y^k$ .

*Proof.* This is just repeated multiplication, so we proceed by induction on  $k$ .

- IH:  $P(k) := x^k \equiv_n y^k$
- Base case ( $k = 1$ ): this is the theorem assumption.
- IS: Assume that  $x^{k-1} \equiv_n y^{k-1}$ . Then

$$\begin{aligned}
 x^k &\equiv_n x^{k-1} \cdot x \\
 &\equiv_n y^{k-1} \cdot x && \text{(previous theorem, taking } a = x^{k-1}, b = y^{k-1}, \text{ and } c = x) \\
 &\equiv_n x \cdot y^{k-1} && \text{(commutativity)} \\
 &\equiv_n y \cdot y^{k-1} && \text{(previous theorem, taking } a = x, b = y, \text{ and } c = y^{k-1}) \\
 &\equiv_n y^k
 \end{aligned}$$

- By induction, for all  $k \geq 1$ ,  $x^k \equiv_n y^k$ .

□

**Warning:** The same is not true for the exponent  $k$ . E.g.,  $1 \equiv_5 6$ , but  $2^1 \not\equiv_5 2^6$  (they have remainders 2 and 4, respectively).

Let's see an example: What are the last two digits of

$$x := 11335^{1111}(6 + 7799^{5000})?$$

That's the same as asking for  $x \bmod 100$ .

General strategy: replace intermediate calculations with their remainders, as early and often as we can. This helps us work with smaller numbers.

First of all,

$$x \equiv_{100} 35^{1111}(6 + 99^{5000}).$$

(Not allowed to just reduce the exponents mod 100.) For the right exponent,  $99 \equiv_{100} -1$ , so  $99^{5000} \equiv_{100} (-1)^{5000} \equiv_{100} 1$ . For the left term, look for a pattern:

$$\begin{aligned} 35^1 &\equiv_{100} 35 \\ 35^2 &\equiv_{100} 25 \\ 35^3 &\equiv_{100} 25 \cdot 35 \equiv_{100} 75 \\ 35^4 &\equiv_{100} 75 \cdot 35 \equiv_{100} 25. \end{aligned}$$

Will continue bouncing between 25 and 75. So  $35^{1111} \equiv_{100} 75$ . We find  $x \equiv_{100} 75 \cdot (6+1) \equiv_{100} 25$ , so this must be the remainder.

## 5 Division

Addition, Subtraction, Multiplication, and *bases* of exponents can be substituted mod  $n$  (but not the exponents).

Can we divide mod  $n$ ? Suppose  $3x \equiv_6 3$ . Can we “divide both sides by 3” and conclude that  $x \equiv_6 1$ ? No. (Consider e.g.:  $3 \times 5 \equiv_6 3$ .)

A *multiplicative inverse* of  $x$ , denoted  $x^{-1}$ , is a number you can multiply  $x$  by to get 1. In  $\mathbb{R}$ , the multiplicative inverse of 3 is  $3^{-1} = 1/3$ , because  $3 \cdot 1/3 = 1$ . If “1/3” made sense mod 6, then we could multiply both sides by 1/3 to conclude that  $5 \equiv_6 1$ . So 3 doesn’t have a multiplicative inverse mod 6.

When do mod  $n$  inverses exist for a number  $a$ ?

**Theorem 8.**  $a$  has an inverse mod  $n$  IFF  $\gcd(a, n) = 1$ .

*Proof.*  $a$  has an inverse mod  $n$  IFF exists  $b$  such that  $ab \equiv_n 1$  IFF exists  $b$  and  $q$  such that  $ab - 1 = nq$  (i.e.  $ab - nq = 1$ ) IFF 1 is a linear combination of  $a$  and  $n$  IFF  $\gcd(a, n) = 1$ .  $\square$

**Corollary 9.** If  $p$  is prime and  $a \not\equiv_p 0$ , then  $a$  has an inverse mod  $p$ .

*Proof.*  $\gcd(a, p)$  must be  $p$  (if  $p \mid a$ ) or 1 (only other factor of  $p$ ). Now apply previous result.  $\square$

Having a multiplicative inverse means we “can cancel from both sides” or “divide” by that amount. E.g., 7 and 13 are inverses of each other mod 30. If we know  $7x \equiv_{30} 14$  can we conclude that  $x \equiv_2 \text{ mod } 30$ ? Instead of dividing, let’s multiply both sides by 13:

$$\begin{aligned} 7x &\equiv_{30} 14 \\ 13 \cdot 7x &\equiv_{30} 13 \cdot 14 \\ 91x &\equiv_{30} 182 \\ x &\equiv_{30} 2 \end{aligned}$$

So yes, since 7 has a multiplicative inverse, we can “cancel it from both sides”.

What about  $7x \equiv_{30} 12$ ? This time, we cannot “cancel” in the usual way, but we can still multiply by 13:

$$\begin{aligned} 7x &\equiv_{30} 12 \\ 13 \cdot 7x &\equiv_{30} 13 \cdot 12 \\ 91x &\equiv_{30} 156 \\ x &\equiv_{30} 6 \end{aligned}$$

Important fact:

**Theorem 10** (Fermat’s Little Theorem). *If  $p$  is prime and  $a \not\equiv_p 0$ , then  $a^{p-1} \equiv_p 1$ .*

(Not to be confused with Fermat’s **Last** Theorem. Very different, much harder.)

*Proof.* Idea: look at numbers  $a, 2a, 3a, \dots, (p-1)a$ . Claim this is the same as  $1, 2, 3, \dots, (p-1) \pmod p$ , possibly in jumbled order. E.g.,  $p = 7$ ,  $a = 3$ ,  $3, 6, 9, 12, 15, 18 \equiv_7 3, 6, 2, 5, 1, 4$ .

None are  $0 \pmod p$ , so there are only  $p-1$  possible remainders. Enough to show there are no duplicates.  $ai \equiv_p aj$  implies  $i \equiv_p j$ , because  $a$  has a multiplicative inverse mod  $p$ . No two of the numbers  $1, 2, \dots, p-1$  are equiv mod  $p$ , so no duplicates.

Now, since both sets are same mod  $p$ , their product is congruent mod  $p$ :

$$(p-1)! \cdot a^{p-1} \equiv_p (p-1)!.$$

And since  $\gcd((p-1)!, p) = 1$ , we know  $(p-1)!$  has an inverse mod  $p$ , so we can cancel it:  $a^{p-1} \equiv_p 1$ . Hooray!  $\square$

We saw earlier that we cannot reduce exponents mod  $n$  when doing arithmetic mod  $n$ . However, if  $n$  is prime, FLT gives us a way to reduce exponents anyway; we reduce mod  $n-1$  instead of mod  $n$ .

## 6 Some Simple Applications of Modular Arithmetic

**Theorem 11.** A number is divisible by 9 IFF its sum of digits is divisible by 9.

*Proof.* Say  $n = \sum_{i=0}^k d_i 10^i$ . Note that  $10^i \equiv_9 1^i \equiv_9 1$ , so  $\sum_{i=0}^k d_i 10^i \equiv_9 \sum_{i=0}^k d_i \cdot 1$ .

We get a stronger result!  $\text{rem}(n, 9) = \text{rem}(s(n), 9)$ . The divisibility trick is just checking whether both sides are 0.  $\square$

Another application: ISBN numbers,  $(a_1, \dots, a_{10})$ . Can think of the first 9 digits as the actual number, while the 10th digit is a checksum, where  $a_1 + 2a_2 + 3a_3 + \dots + 10a_{10} \equiv_{11} 0$ . Given first 9 digits, how do we know a 10th digit exists? Because 10 has a multiplicative inverse mod 11. (Note however, that if this last digit should be 10, then the number is not a valid ISBN.)

Can prove that if a single digit gets copied wrong, the check won't come out to 0 mod 11. Similarly, if two adjacent unequal digits are swapped, check won't come out to 0 mod 11.

Similar ideas are used in other error-correcting scenarios, e.g., redundant memory storage, RAID. A simple hypothetical strategy: have first two disks store bits  $b_1$  and  $b_2$ , while the third disk stores  $b_3 := (b_1 \oplus b_2)$ . If 2nd disk fails, can recover  $b_2$  as  $b_1 \oplus b_3$ . ( $\oplus$  denotes addition mod 2, or parity.)

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